

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Equations — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.3: Quadratic Equations
Driving Question	DRIVING QUESTION: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring the Phenomenon: The Githurai Harambee Hall Problem	Students observed a hand-drawn sketch of the Githurai Harambee hall — a rectangle with width w , length $(w + 2)$, and Area = 120 m^2 — and attempted to find w using trial-and-error arithmetic, generating a whole-class table of (w, Area) pairs.	Students learned that multiplying two expressions each containing w produces a w^2 term, making the equation $w^2 + 2w - 120 = 0$ structurally different from linear equations; they identified this as a quadratic equation in the general form $ax^2 + bx + c = 0$ with $a = 1$, $b = 2$, $c = -120$.	Students explained that linear methods fail for the hall problem because the area relationship forces w to be multiplied by itself, creating a squared term that cannot be isolated by simple arithmetic or one-step inverse operations — and that a new class of mathematical tools (quadratic equation methods) is needed to find the exact dimensions for the community's hall.
Lesson 2 Expanding and Simplifying Quadratic Expressions	We observed that multiplying $(w)(w + 2)$ from the Githurai hall sketch and $(x + 5)(x + 3)$ from the Kericho farm both produced expressions with a squared term, a middle term, and a constant — even when different methods were used.	We learned that the distributive law and FOIL method are systematic ways to expand the product of any two linear expressions, always producing a quadratic expression of the form $ax^2 + bx + c$, and that the Harambee hall problem can now be written precisely as $w^2 + 2w - 120 = 0$.	We can explain that the squared term in a quadratic expression arises because the variable appears in both brackets and multiplies itself during expansion; this is why the hall problem cannot be solved with a linear equation and why the committee in Githurai needs quadratic mathematics to find the dimensions of their hall.
Lesson 3 Factorisation of Quadratic Expressions	Learners observed three teacher-narrated worked examples of factorisation: (1) $x^2 + 5x + 6$ (Kericho tea farmer profit), (2) $w^2 + 2w - 120$ (the Harambee hall equation), and (3) $x^2 - 49$ (Mombasa tiling contractor — difference of two squares). They completed a 6-question individual worksheet applying	Learners learned that: (a) factorisation is the reverse of expansion; (b) to factorise $x^2 + bx + c$, find two numbers that multiply to c and add to b ; (c) to factorise $ax^2 + bx + c$, use the splitting-the-middle-term method (multiply $a \times c$, find factor pair, split the middle term, factor by grouping); (d) perfect	Learners explained their factorisation reasoning through three group problems: (A) the hall equation — factorised and interpreted, rejecting the negative root; (B) Mama Akinyi's Gikomba mandazi revenue model $6n^2 + 11n - 10$, factorised to $(2n + 5)(3n - 2)$; (C) the Kisumu garden perfect

	these patterns.	square trinomials and difference of two squares follow special patterns; (e) the Harambee hall equation $w^2 + 2w - 120 = 0$ factorises to $(w + 12)(w - 10) = 0$, giving real width $w = 10$ m and length 12 m; (f) negative roots are rejected when the variable represents a physical measurement.	square $x^2 + 14x + 49 = (x + 7)^2$. They updated their cumulative hall model diagrams and posted DQB sticky notes anchored to the insight: 'Factorisation lets us undo expansion to find hidden values.'
Lesson 4 Solving Quadratic Equations by Factorisation	We observed that setting each factor of $(w + 12)(w - 10) = 0$ equal to zero gives $w = -12$ and $w = 10$. When we substituted both values back, both made the equation true. We also observed that in the Githurai hall problem, only $w = 10$ makes sense as a physical width.	We learned the Zero Product Property: if two factors multiply to give zero, at least one factor must equal zero. We learned that a quadratic equation solved by factorisation always produces two solutions, and that context — not just algebra — determines which solution to accept.	We explained that the Harambee hall in Githurai must be 10 m wide and 12 m long because the Zero Product Property gives us two roots ($w = 10$ and $w = -12$) but a negative width is physically impossible. We also explained that sometimes both roots are positive and we must use deeper contextual reasoning to choose between them.
Lesson 5 Solving Quadratic Equations by Completing the Square	Learners observed that the Nakuru school garden equation ($x^2 + 6x = 48$) could not be solved by factorisation over integers. Through a geometric grid-paper construction, they saw a literal square being 'completed' by adding a 3×3 corner piece, and observed that the resulting diagram corresponded exactly to the algebraic identity $(x + 3)^2 = x^2 + 6x + 9$.	Learners learned that any quadratic equation of the form $x^2 + bx = c$ can be rewritten as $(x + b/2)^2 = c + (b/2)^2$ — a process called completing the square. They learned that this method always works, produces exact (sometimes irrational) solutions, and gives the same answer as factorisation when both methods apply. They applied this to the Nakuru garden and re-verified the Githurai hall dimensions.	Learners explained why completing the square works by connecting each algebraic step to a region in their geometric diagram. They explained that the Githurai committee can now use this method for any rectangular-area planning problem — even when the dimensions do not produce neat integer factor pairs — and updated the DQB with the headline: 'Completing the square ALWAYS works, even when factorisation fails.'
Lesson 6 The Quadratic Formula & The Mombasa Stone Problem	Students observed the algebraic derivation of the quadratic formula by completing the square on the general form $ax^2 + bx + c = 0$, step by step using a Derivation Scaffold card. They also observed two applied Kenyan scenarios: (1) a stone thrown from a Mombasa cliff modelled by $h(t) = -5t^2 + 20t + 25 = 0$, and (2) a mandazi vendor's break-even pricing modelled by $-2x^2 + 40x - 150 = 0$. Students observed that one of the Mombasa solutions ($t = -1$) had to be rejected because time cannot be negative.	Students learned that: (1) the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$ can be derived by completing the square on any quadratic equation in standard form; (2) the discriminant $\Delta = b^2 - 4ac$ determines the number of real solutions before full solving ($\Delta > 0$ gives two solutions, $\Delta = 0$ gives one, $\Delta < 0$ gives none); (3) the quadratic formula works for every quadratic equation, even those that cannot be factorised easily; (4) mathematical solutions must always be interpreted in context — not all algebraic solutions are physically or logically meaningful.	Students explained, using the Claim-Evidence-Reasoning strategy and their method comparison table, when each of the three methods (factorisation, completing the square, quadratic formula) is most appropriate. They explained why $t = -1$ is rejected in the Mombasa problem and why the quadratic formula is described as a 'universal tool'. They also explained, using the discriminant, why equation C ($x^2 + x + 3 = 0$) from the Predict phase has no real solutions, revisiting and correcting their initial predictions.
Lesson 7 Application, Model	We observed that the Githurai Harambee committee could not find the hall	We learned that factorisation is most efficient when the equation has integer	We explained that the three solution methods are not competing alternatives but

Building and Consolidation — Closing the Harambee Hall Story	dimensions using arithmetic or linear equations — but when we wrote $w(w + 2) = 120$ and rearranged it to $w^2 + 2w - 120 = 0$, three different methods — factorisation, completing the square, and the quadratic formula — all gave us $w = 10$ and $w = -12$. We also observed that new community problems about fish-drying yards, tiled borders, and chapati pricing could all be modelled and solved as quadratic equations.	factor pairs; completing the square always works and reveals the geometric meaning of the solution; the quadratic formula is universal and essential when the discriminant is not an obvious perfect square. We learned that both algebraic solutions must always be interpreted in context — a negative width is mathematically valid but physically meaningless. Most importantly, we learned that a quadratic equation is a precise tool for answering real planning questions that a community cannot solve by guessing.	complementary tools — each suited to different problem structures. We explained the full unit journey: quadratic expressions emerge from area and revenue contexts; expansion reveals their structure; factorisation reverses expansion to find roots; completing the square and the quadratic formula extend our reach to equations that resist factorisation. We explained to a community audience — through our models and reflections — that knowing how to write and solve a quadratic equation gives any person the power to answer real questions about land, construction, pricing, and planning in their own community.
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